Alegre, Jhon Isaac S.
Prelim Quiz #1

Your goal is to write a playbook named `gateway.yml` that prepares a server to handle high-traffic file requests.

Specifics:

1. Use a single task with a loop to install nginx, redis-server, and ufw (Uncomplicated Firewall).

```
- hosts: all

 become: true

 tasks:





 tasks:

 - name: Install Nginx

   ansible.builtin.apt:

     name: nginx

     state: present

   become: yes



 - name: Install redis-server

   ansible.builtin.apt:

     name: redis-server

     state: present

   become: yes
```

```yaml
- name: Install UFW
  ansible.builtin.apt:
    name: ufw
    state: present
  become: yes


- name: Create user directories inside target user's home
  ansible.builtin.file:
    path: "~/{{ item }}"
    state: directory
    mode: '0755'
  loop:
    - test_d1
    - test_d2
    - test_d3
  become: yes
```

**Results:**

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                bash

isaac@Workstation:~/ansible$ ansible-playbook gateway.yml -K
BECOME password:
[WARNING]: While constructing a mapping from /home/isaac/ansible/gateway.yml, line 1, column 3, found
a duplicate dict key (tasks). Using last defined value only.

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [Install Nginx] ***********************************************************************************
ok: [192.168.56.112]

TASK [Install redis-server] ****************************************************************************
ok: [192.168.56.112]

TASK [Install UFW] *************************************************************************************
ok: [192.168.56.112]

TASK [Create user directories inside target user's home] ***********************************************
ok: [192.168.56.112] => (item=test_d1)
ok: [192.168.56.112] => (item=test_d2)
ok: [192.168.56.112] => (item=test_d3)

PLAY RECAP *********************************************************************************************
192.168.56.112             : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
```

**Verify:**

```
isaac@Workstation:~$ nginx -v
nginx version: nginx/1.18.0 (Ubuntu)
isaac@Workstation:~$ redis-server --version
Redis server v=6.0.16 sha=00000000:0 malloc=jemalloc-5.2.1 bits=64 build=e91d798
6ac6d5bb6
isaac@Workstation:~$ ufw --version
ufw 0.36.1
Copyright 2008-2021 Canonical Ltd.
isaac@Workstation:~$
```

2. Implement the use of roles.

Nginx:

```yaml
---
# tasks file for nginx
- name: Install Nginx
  ansible.builtin.apt:
    name: nginx
    state: present
  become: yes
```

**Redis-server:**

```yaml
---
# tasks file for redis-server
- name: Install redis-server
  ansible.builtin.apt:
    name: redis-server
    state: present
  become: yes
```

**Ufw:**

```yaml
---
# tasks file for ufw
- name: Install UFW
  ansible.builtin.apt:
    name: ufw
    state: present
  become: yes
```

**Result:**

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                bash

isaac@Workstation:~/ansible$ ansible-playbook prelim.yml -K
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Platform linux on host 192.168.56.112 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.112]

TASK [nginx : Install Nginx] ***************************************************************
ok: [192.168.56.112]

TASK [redis-server : Install redis-server] *************************************************
ok: [192.168.56.112]

TASK [ufw : Install UFW] *******************************************************************
ok: [192.168.56.112]

PLAY RECAP *********************************************************************************
192.168.56.112             : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0

isaac@Workstation:~/ansible$
```

**Files:**

```
> nginx
> redis-server
> ufw
  inventory.ini
  prelim.yml
```

You may search the internet, however no using of ChatGPT, Gemini, or the like.

You may also use your previous activities as reference.